

# Activity day – 8

1. Our very first objective is to write c program of finding height and slant height of a very cone when radius and volume are given .

So, here is our c program to achieve this

```
#include <stdio.h>
#include <math.h>
int main()
{
    double h, l, s, V, r;

    printf("Enter the volume and radius:");
    scanf("%lf %lf\n", &V, &r);

    h = (3*V) / (3.14 * r * r);
    l = sqrt(r*r + h*h);
    s = 3.14 * r * l + 3.14 * r * r;

    printf("Height, slant height, surface area of cone are: %lf, %lf, %lf\n", h, l, s);
    return 0;
}
```

## Test case to verify our program

| Test case | Input(s)                | Expected output(s)                                          | Actual Output(s)                                               | Result (pass or fail) |  |
|-----------|-------------------------|-------------------------------------------------------------|----------------------------------------------------------------|-----------------------|--|
| 1.        | Radius=7,<br>Volume 120 | Height =2.33<br>Slant height =7.38<br>Surface area = 316    | Height =2.339<br>Slant height =7.381<br>Surface area = 316.12  | PASS                  |  |
| 2.        | Radius=6,<br>Volume 128 | Height =3.39<br>Slant height =6.89<br>Surface area = 242.94 | Height =3.391<br>Slant height =6.894<br>Surface area = 242.942 | PASS                  |  |

|    |                            |                                                                            |                                                                               |      |  |
|----|----------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------------------|------|--|
| 3. | Radius=9,<br>Volume<br>130 | Height<br>=1.53<br>Slant<br>height<br>=9.12<br>Surface<br>area =<br>512.34 | Height<br>=1.534<br>Slant<br>height<br>=9.128<br>Surface<br>area =<br>512.348 | PASS |  |
|    |                            |                                                                            |                                                                               |      |  |
|    |                            |                                                                            |                                                                               |      |  |

Algorithm:

1.START

2.take the input r, the radius, volume , v,  
from the user.

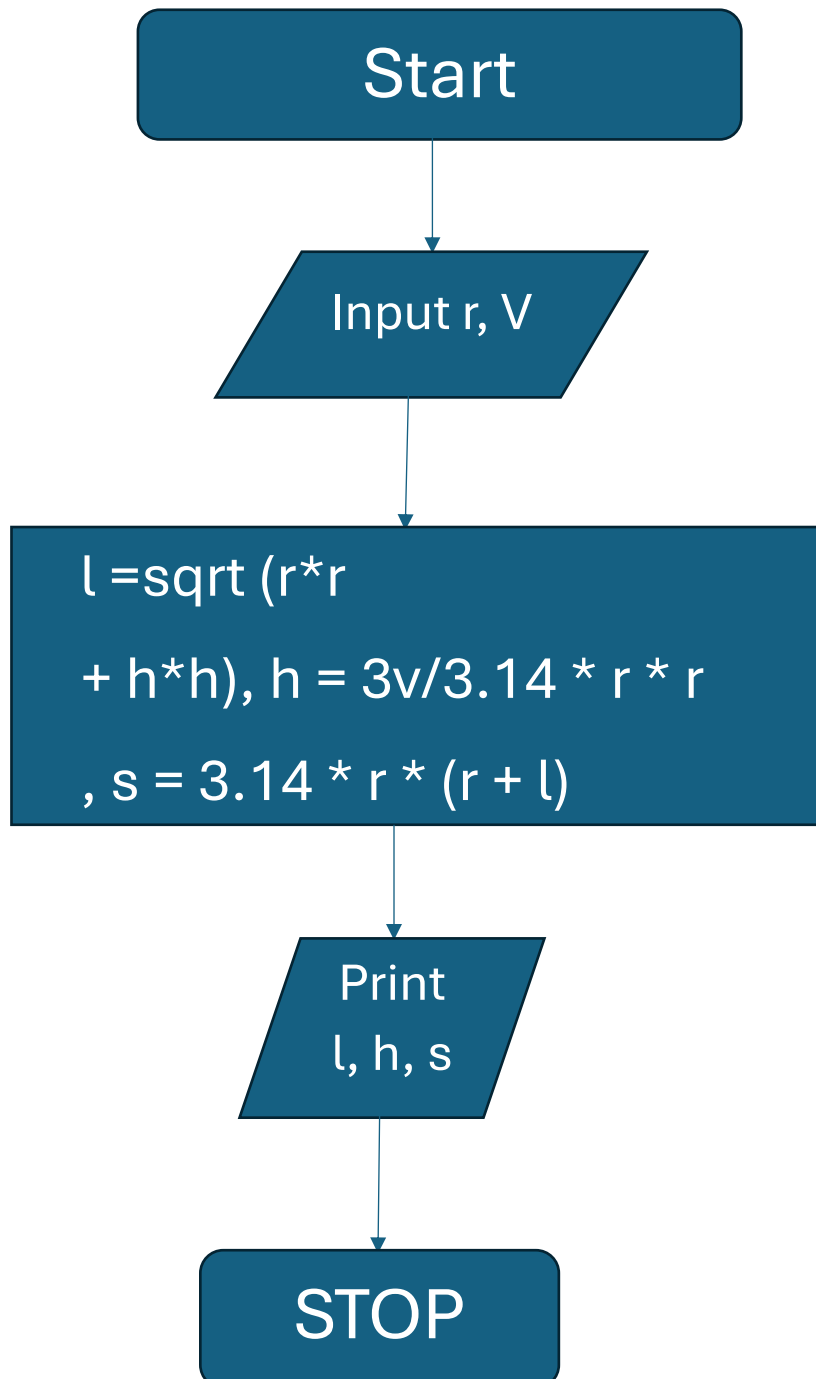
3. calculate height ,  $h = 3v/3.14 * r * r$

4. calculate slant height ,  $l = \text{sqrt}(r*r + h*h)$

5. calculate surface area,  $s = 3.14 * r * ( r + l )$

6. print h, l, s

## 7. STOP



Our next task is to code a c program to find the perimeter of a quadrilateral whose 4 vertices are given

Here is our program

```
#include <stdio.h>
#include <math.h>

double distance(double x1, double x2, double y1, double y2)
{
    return sqrt(pow(x2-x1, 2) + pow(y2-y1, 2));
}

double main()
{
    double x1, x2, x3, x4, y1, y2, y3, y4;
    double first, second, third, fourth, total;

    printf("Enter the first set of vertices:");
    scanf("%lf %lf", &x1, &y1);

    printf("Enter the second set of vertices:");
    scanf("%lf %lf", &x2, &y2);

    printf("Enter the third set of vertices:");
    scanf("%lf %lf", &x3, &y3);

    printf("Enter the fourth set of vertices:");
    scanf("%lf %lf", &x4, &y4);

    first = distance(x1, y1, x2, y2);
    second = distance(x2, y2, x3, y3);
    third = distance(x4, y4, x3, y3);
    fourth = distance(x1, y1, x4, y4);

    total = first + second + third + fourth;

    printf("Total perimeter: %.2lf\n", total);
    return 0;
}
```

Compilation

```
(kali@kali)-[~]
$ vi newd.c

(kali@kali)-[~]
$ gcc newd.c -o newd -lm
```

## Algorithm

1. START
2. Define a distance function to calculate distance between two coordinates.
3. Declare double types  $x_1, x_2, x_3, x_4, y_1, y_2, y_3, y_4$ .
4. Calculate the distance between each set of coordinates expect with diagonally related.
5. Add them and make those equal to a variable perimeter.
6. Print perimeter
7. STOP

| Test cases | INPUT                            | OUTPUT | Expected Output | PASS /FAIL |
|------------|----------------------------------|--------|-----------------|------------|
| 1          | X = 2, 4, 6, 5<br>Y = 3, 5, 7, 8 | 9.15   | 9.17            | PASS       |
| 2          | X = 6, 3, 4, 1<br>Y = 9, 9, 5, 7 | 25.58  | 25.6            | PASS       |
| 3          | x = 5, 3, 1, 7<br>y = 9, 7, 2, 4 | 17.94  | 17.99           | PASS       |
|            |                                  |        |                 |            |

# FLOWCHART

